
CS246: Database Management Systems Lab

Week # 07 (1 Questions, 102 Marks)

Lab session: AL1

Held on: Monday, 23-Feb-2026 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 3

Submission dead line: 17:00 hrs, 23-Feb-2026

Submissions accepted from: 16:45 hrs, 23-Feb-2026

Submissions accepted till: 17:30 hrs, 23-Feb-2026

Every one minute delay in submission will result in one mark deduction

Please do not quote Network issues or technical issues after 17:30hrs

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- This lab assignment is based on the concepts covered in chapter 3 introduction to SQL in the previous week CS245 theory class.
- Following manual files are provided along with this lab problem:
 1. SELECT manual pages documentation
 2. LOAD DATA manual pages documentation

Question 1: (102 points)

Using MySQL perform the following tasks:

Task 01 (3 Marks) Create a database named **week07**. Perform the following tasks using this database.

Task 02 Create the following tables

1. (3 marks) A table **student** containing the following

1st column	cid	string of characters of fixed size 7	Read the note below
2nd column	roll_number	string of characters of fixed size 10	Read the note below
3rd column	name	string of characters of fixed size 100	NOT NULL
4th column	approval_status	string of characters of fixed size 20	NOT NULL
5th column	credit_status	string of characters of fixed size 10	NOT NULL

Note: • roll_number and cid as primary key.

2. (3 marks) A table **course** containing the following

1st column	cid	string of characters of fixed size 7	primary key
2nd column	name	string of characters of fixed size 100	NOT NULL

3. (3 marks) A table **credit** containing the following

1st column	cid	string of characters of fixed size 7	primary key
2nd column	l (lecture)	integer	NOT NULL
3rd column	t (tutorial)	integer	NOT NULL
4th column	p (practical)	integer	NOT NULL
5th column	c (credits)	float	NOT NULL

Task 03 (15 marks) Populate the data using **load data** statement (manual page corresponding to load data is shared). In case you have issues using load data statement, write several **INSERT** statements and populate the data using C++ program.

1. (5 marks) populate data from the file **students-credits.csv** into table **student**
2. (5 marks) populate data from the file **courses.csv** into table **course**
3. (5 marks) populate data from the file **credits.csv** into table **credit**

Task 04 Queries involving single table

1. (5 marks) List all the columns of the **student** table pertaining to Adarsh Kumar Udai
2. (5 marks) List cid, name and credit_status of **student** whose credit_status is **Credit** and cid is **EE 390**
3. (5 marks) List the columns cid, roll_number, credit_status, approval_status of **student** whose approval_status is 'Pending' and credit_status is 'Credit'
4. (5 marks) List cid, l, t, p, c of **course** table in which number of credits c is not 6
5. (5 marks) List student roll_number, name, cid, credit_status and approval_status where credit_status is 'Audit' and approval_status is 'Approved'

Task 05 Queries involving multiple tables

1. (5 marks) List course name, l, t, p, c having course credits 8.
2. (5 marks) List course name, l, t, p, c with a course having tutorial component (that **t** > 0)
3. (5 marks) List course cid, name, l, t, p, c of those courses having 6 credits and not having credit structure 3-0-0 (correspond to l-t-p)
4. (5 marks) List the cid, course name, student name, l, t, p, c of each course registered by student named **Pasch Paul 01e**
5. (5 marks) List the roll_number, student name, cid, course name, l, t, p, c of students whose credit_status is 'Credit' and credit structure is 3-1-0-8 and course offered by ECE department (cid starts with EE).

Task 06 String operations

1. (5 marks) List cid and student name in which 'ATUL' appears anywhere in the student name. Use the function **UPPER** to convert a string to all upper case letters

2. (5 marks) List roll_number, credit_status and course name of student whose course name starts with “introduction to”. Convert all the characters in course name to lower letters using function LOWER
3. (5 marks) List all third year courses cid’s. Third year courses are identified by examining 4th character of cid. It should be 3.
4. (5 marks) List the cid and course name of minor courses offered by the institute. Minor courses have course codes where the 5th character is ‘2’ and end with ‘M’ (e.g., CC 123M, CC 724M, CC 829M). Assume course codes include a space after the department code.
5. (5 marks) List student name, cid, course name where credit_status is ‘Credit’ and student name starts with “A” ends with “TA”. First convert all the characters to capital letters using function UPPER.

Submission Procedure Upload the .sql file in MS assignments site.